EXPLORATORY DATA ANALYSIS (EDA) SUMMARY

MedBhasha — A Bengali Clinical NLP Hackathon (Round 1: Onboarding & Annotation)

Submitted by: Team Pragyan | Corpus File: Pragyan_Round1_MedBhasha_Annotated.csv | Validator Status: PASS (0 Errors)

Total Clinical Notes	Total Tokens	Unique Vocabulary	Labeled Entity Tokens	Outside ('O') Tokens
762 Notes (n001–n762)	9,172 Tokens	1,313 Unique Words	2,727 Tokens (29.73%)	6,445 Tokens (70.27%)

1. Plain-English Overview of Our Bengali Clinical Corpus

To train a reliable computer model that can read and understand medical information in Bengali, we first needed to build a clean, well-organized collection of clinical sentences and label the medical words inside them. Our final Round 1 dataset contains 762 separate clinical notes made up of 9,172 individual words and punctuation marks called 'tokens'. Every single note was gathered strictly from competition approved open sources and synthetic generation which ensure that no real hospital patient records or personal identities were ever used.

1.1 How Our 762 Clinical Notes Were Gathered

Source Category	Specific Source / Method	Notes Count	What Kind of Text It Adds to Our Dataset
Open Public Health Portals	Vikaspedia Health (Bengali)	54	Verified disease overviews, screening advice (PSA, dialysis, epilepsy, kidney failure) and preventive care.
Open Public Health Portals	Shohay Health (Bengali)	10	Pediatric care guidelines, fever medication instructions, infant feeding rules and digestive/ear conditions.
Synthetic Clinical Notes	LLM-Generated Clinical Scenarios	40	Doctor-style clinical documentation packed with lab tests, exact test numbers, multi-word drug dosages and surgeries.
Patient Complaint Texts	De-identified / Adapted Complaints	658	Everyday spoken Bengali describing how patients naturally explain pain, swelling, fever and discomfort to a doctor.

1.2 How the BIO Tagging Split Works

In our final CSV file, every word and punctuation mark sits on its own row with a tag from the official 10-category schema. Out of 9,172 total rows:

- **2,162 Beginning ('B-') Tokens:** It mark the first word of a medical term, for example, '□□□□□□' in '□□□□□□ □□□□□□' or words like '□□□□'.
- **565 Inside ('I-') Continuation Tokens:** It mark the second, third, or fourth word of a multi-word medical phrase, for example, '□□□□□□' in '□□□□□□ □□□□□□' or '□□□□□□ □□□□ □□□□□□' in a dosage span.
- **6,445 Outside ('O') Tokens (70.27% of the dataset):** It represent normal Bengali grammar words, pronouns, verbs ('□□□□', '□□□', '□□□□□'), and punctuation marks ('□', ','). It is having roughly 70% 'O' tokens is completely natural and essential so that our Round 2 model learns to tell the difference between ordinary grammar words and true clinical entities.

2. Note-Length Distribution

Before training a language model, we need to check whether our notes are very short fragments, medium-sized sentences or long paragraphs. We analyzed the number of tokens per note across all 762 notes using both a Histogram and a Boxplot.

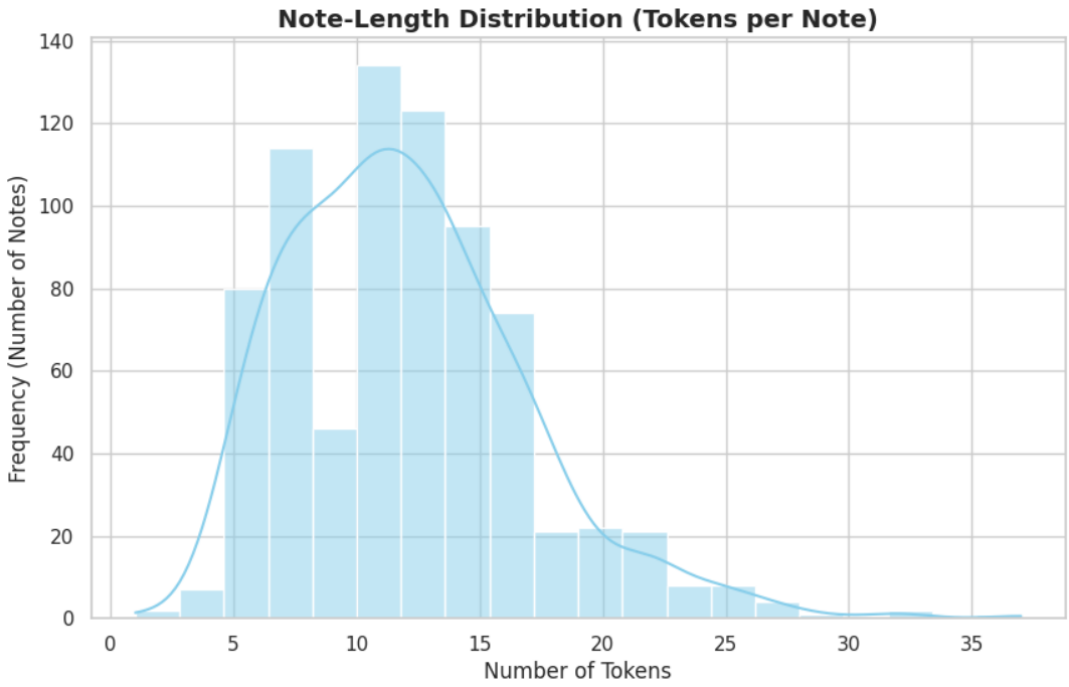


Figure 1: Histogram of Note-Length Distribution (Number of Tokens per Clinical Note across all 762 Notes).

2.1 What the Histogram Tells Us

- **Most Notes Are Concise (6 to 16 Tokens):** As seen in Figure 1, the tallest sky-blue bars sit right in the middle between 6 and 16 tokens which peak at 11 to 13 tokens where more than 130 notes cluster together. In everyday terms, when a patient describes a health problem in Bengali they almost always state it in one clear sentence of around 10 to 14 words.
- **Smooth Right-Side Tail (18 to 37 Tokens):** The smooth blue curve gradually stretches out to the right up to 37 tokens. This right-hand tail comes from our longer public health guidelines and synthetic doctor notes which contain multiple clinical clauses and punctuation marks.

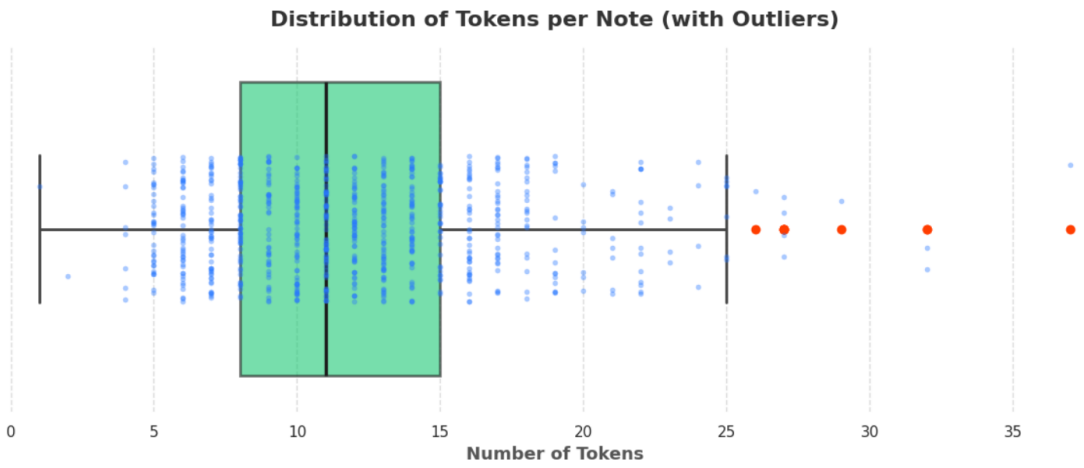


Figure 2: Boxplot and Strip-Plot Overlay of Tokens per Note Highlight the Median, Spread, and Outliers.

2.2 What the Boxplot Tell Us

Total Notes	Mean Length	Median (Q2)	Std. Dev.	Min Tokens	25th % (Q1)	75th % (Q3)	Max (Outliers)
762	12.04	11.00	4.86	2	8.00	15.00	37 (9 notes > 25)

- **The Green Box:** The green box spans from 8 tokens (25th percentile) to 15 tokens (75th percentile) with the thick black median line sitting right at 11 tokens. Our average note length is 12.04 tokens with a standard deviation of 4.86 tokens.
- **Blue Dots:** Each semi-transparent blue dot represents an actual clinical note. By looking at them distributed smoothly across every vertical column from 2 to 25 confirms that our notes vary naturally in length rather than being generated from a repetitive copy paste template.
- **Orange Dots:** Here, any note longer than 25.5 tokens that means above the right whisker is flagged as a statistical outlier in bright orange at 26, 27, 29, 32, and 37 tokens. These are not data errors, they are comprehensive clinical notes such as renal failure treatment pathways discussion of both dialysis and kidney transplantation in a single note. It keep them ensures our model can handle long and multi-clause medical sentences.

3. Entity Frequency Distribution

Next, we examined how our 2,727 labeled medical tokens comprising of 2,162 distinct entity spans are distributed across the 10 official hackathon categories. By checking it helps us see what our dataset is strongest at and whether any category is missing.

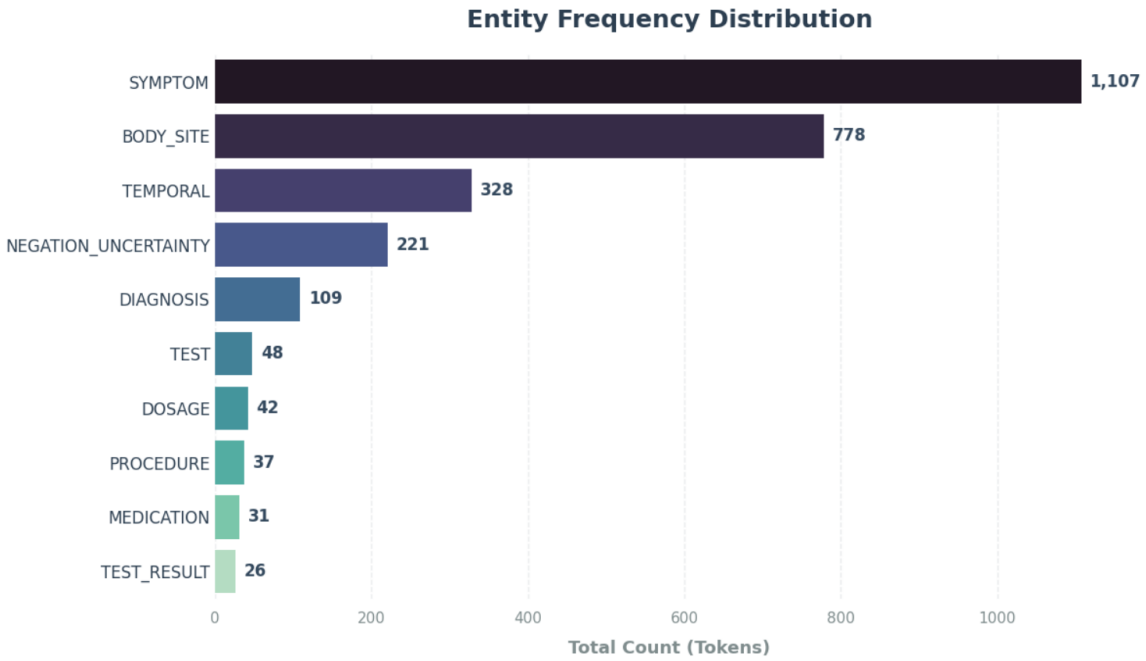


Figure 3: Horizontal Bar Chart of Entity Frequency Distribution (Total Annotated Tokens per Category).

3.1 Complete Breakdown of All 10 Entity Categories

Entity Category	Total Tokens (B+I)	Distinct Spans (B-)	Continuation (I-)	Avg. Words / Span	Common Examples from Our Bengali Corpus
SYMPTOM	1,107 (40.59%)	922	185	1.20	□□□□□, □□□□, □□□□, □□□, □□□□□□□□□, □□□□□□□, □□□□□□, □□□□□ □□□□
BODY_SITE	778 (28.53%)	672	106	1.16	□□□□, □□□□, □□□□, □□□, □□□□, □□□□□, □□□□□ □□□□□□, □□□□□□□□□□ □□□□
TEMPORAL	328 (12.03%)	173	155	1.90	□□ □□□ □□□ □□□, □□□□□ □□ □□□□□□ □□□, □□□□, □□□□, □□□□□□□□
NEGATION_UNCERTAINTY	221 (8.10%)	221	0	1.00	□□, □□□, □□□, □□, □□□□□□, □□□□□□, □□□□□□
DIAGNOSIS	109 (4.00%)	77	32	1.42	□□□□□□□□□□, □□□□□□, □□□□, □□□□□□ □□□□□ □□□□□□, □□□□□□ □□□□□□

TEST	48 (1.76%)	28	20	1.71	□□□□□, □□□□ □□□□□□□□□, □□□□-□□, □□□□□, □□□□□□□□□□□□□□□□, □□□□ □□□□□□□
DOSAGE	42 (1.54%)	11	31	3.82	□□□ □□□□□□, □□□□ □□□□□□, □□ □□□□□□ □□□□□□□□ □□□□□ □□□□ □□□□
PROCEDURE	37 (1.36%)	21	16	1.76	□□□□□□□, □□□□□□□□□□, □□□□□□□□□□, □□□□□□□□□□□□□□□ □□□□□□□□, □□□□□□□□
MEDICATION	31 (1.14%)	25	6	1.24	□□□□□□□□□□□□, □□□□□□□, □□□□□□□□□□□□□□, □□□□□□□□□□□□□□□, □□□□□□□□□□□□□□□□
TEST_RESULT	26 (0.95%)	12	14	2.17	□□□□□□□□□, □□□ □□□□□□, □□□, □□□/□□□ □□□□□□□□□ □□□□□□□□, □□□□□□□□ □□□□□□□□
TOTAL MEDICAL ENTITIES	**2,727 (100%)**	**2,162 Spans**	**565 Tokens**	**1.26**	**All 10 schema categories actively represented (0 missing types)**

3.2 Key Takeaways from the Bar Chart

- **Why Symptoms & Body Sites Lead the Chart:** Together, SYMPTOM and BODY_SITE account for 69.12% of all labeled entity tokens. It is because the largest portion of our corpus consists of primary-care patient complaints where speakers naturally focus on what they are feeling such as pain, fever, itching or vomiting and where on their body it hurts such as head, chest, eyes, teeth or stomach.
- **High Presence of Timeframes & Negations:** Patients frequently describe how long a symptom has lasted ('□□□□□ □□ □□□□□□ □□□') and what functions they cannot perform or symptoms they do not have ('□□□□ □□□□ □□□ □□ □□', '□□□□□□□□□ □□□'). Notably, NEGATION_UNCERTAINTY has 221 B- spans and 0 I- tokens, average length = 1.00 token which proves that our team strictly followed the guideline to tag only the negation word itself without accidentally highlighting the surrounding symptom.
- **Why Dosage Has the Longest Span Length:** Even though DOSAGE has 11 distinct B- spans, it accounts for 42 total tokens because medical dosage instructions in Bengali naturally span several words, for example- '□□ □□□□□□ □□□□□□□□ □□□□□ □□□□ □□□□' is 1 B-DOSAGE token followed by 5 I-DOSAGE tokens.
- **Complete Coverage of Clinical Action Categories:** DIAGNOSIS (109 tokens), TEST (48 tokens), DOSAGE (42 tokens), PROCEDURE (37 tokens), MEDICATION (31 tokens) and TEST_RESULT (26 tokens) are all present with zero missing categories and give our Round 2 NER pipeline clean training examples for every single entity class.

4. Co-occurrence Analysis

In real medical language, clinical terms rarely appear in isolation, they work together in logical pairs inside the same sentence. To fulfill the co-occurrence requirement, we analyzed both how our top two categories- SYMPTOM and BODY_SITE scale together inside individual notes and how all 10 categories pair up across the 762 notes.

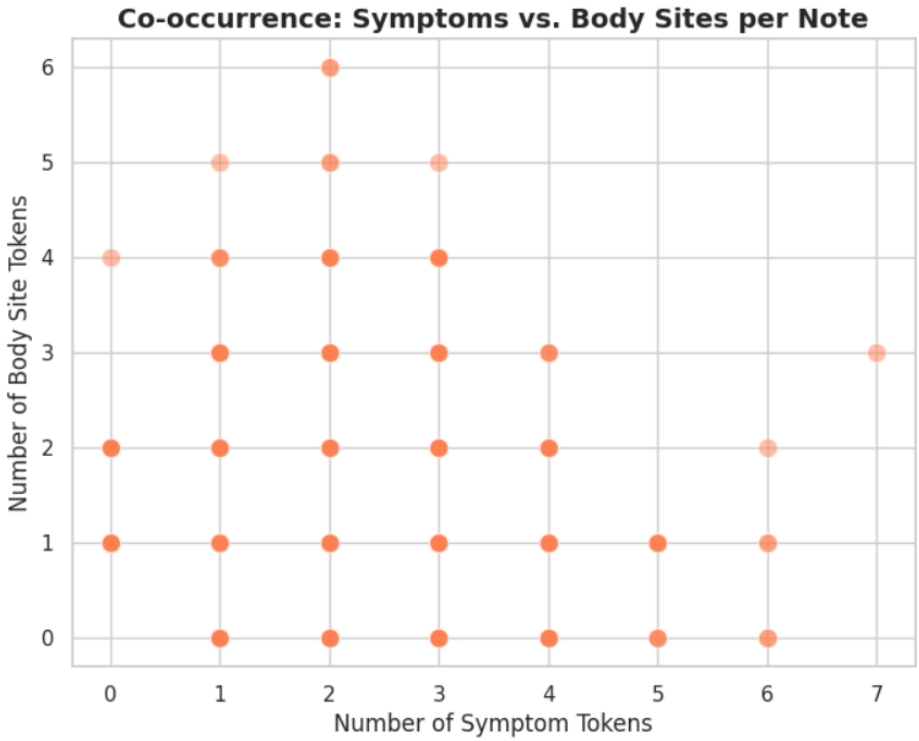


Figure 4: Scatter Plot of Co-occurrence Between Symptom Tokens and Body Site Tokens per Clinical Note.

4.1 What the Scatter Plot Tells Us

- **Dense Cluster at 1–3 Symptoms & 1–3 Body Sites:** In Figure 4, each orange dot shows the number of SYMPTOM tokens (horizontal axis) and BODY_SITE tokens (vertical axis) inside a single note where darker orange circles indicate many notes stacked on the exact same coordinate. The darkest concentration sits right around 1 to 3 Symptoms or 1 to 3 Body Sites with a positive Pearson correlation of r = 0.251.
- **Almost Every Physical Complaint Pairs Anatomy with a Symptom:** Out of 762 total notes, 452 notes (59.32%) contain both a BODY_SITE and a SYMPTOM in the very same note. In simple terms, whenever someone mentions a symptom like pain ('□□□□□'), burning ('□□□□□□ □□□□') or swelling ('□□□□'), they almost always anchor it to a specific body part such as '□□□□ □□□□□', '□□□ □□□□ □□□', '□□□□□ □□□□□□ □□□□□'.
- **Multi-Symptom Notes Scale Up to 7 Symptom Tokens:** Notes on the right side of the scatter plot that means 4 to 7 symptom tokens represent patients describing a chain of connected symptoms across multiple body parts such as chest pain accompanied by sweating, cold hands and feet and shortness of breath.

4.2 Top Note-Level Entity Co-occurrence Pairs Across the Corpus

Entity Pair Appearing in the Same Note	Notes Count	% of All 762 Notes	Real Clinical Pattern & Example from Our Corpus
BODY_SITE + SYMPTOM	452 Notes	59.32%	Anatomical site followed immediately by complaint (e.g., '□□□□ □□□□□', '□□□□ □□□□□', '□□□□□ □□□□ □□□□').
NEGATION_UNCERTAINTY + SYMPTOM	118 Notes	15.49%	Ruling out a symptom or stating inability (e.g., '□□□□□□□□□ □□ □□□□□ □□□□ □□□□□□ □□□', '□□□□ □□□ □□ □□').
BODY_SITE + NEGATION_UNCERTAINTY	113 Notes	14.83%	Confirming no abnormality in a body organ (e.g., '□□□□□□□ □□□□ □□□□ □□□ □□□□□□', '□□□□□ □□□□ □□ □□□□□').
SYMPTOM + TEMPORAL	112 Notes	14.70%	Duration or onset of a symptom (e.g., '□□ □□□□ □□□ □□□ □□□□□□ □□□□ □□□□', '□□□□□ □□□ □□□□□□ □□□ □□□□□□□ □□□□').
BODY_SITE + TEMPORAL	90 Notes	11.81%	How long a specific body part has been affected (e.g., '□□□□□ □□ □□□ □□□ □□□□ □□□□□ □□□□ □□□□ □□').
BODY_SITE + DIAGNOSIS	38 Notes	4.99%	Linking an organ to a named disease (e.g., '□□□□□□ □□□□□ □□□□□ □□□□□ — □□□□□□ □□□□□', '□□□□□□□ □□□□□□').
DIAGNOSIS + SYMPTOM	31 Notes	4.07%	Connecting a disease to its outward signs (e.g., '□□□□□□□□□□ □□□□□ □□□□□□ □□□□□□□□□ □ □□□ □□□ □□□').
MEDICATION + DOSAGE	11 Notes	1.44%	Drug name followed directly by strength and schedule (e.g., '□□□□□□□□□□□□□ □ □□□□□□ □□□ □□□□□ □□□□□').

5. Annotation Quality Self-Check & Round 2 Readiness

As highlighted in the hackathon rules, performing this Exploratory Data Analysis served as our internal quality check before final submission:

- **Zero Structural or Schema Errors:** We verified our final annotated corpus against the official validate.py script under the BIO scheme. It passed with 0 errors and 0 warnings across all 762 notes and 9,172 tokens which confirm no orphan 'I-' tags, no missing tokens and exact casing for all 10 entity categories.
- **Natural Spelling Variations & Code-Mixing Preserved:** In accordance with Section 6 of the Annotation Guidelines, we did not artificially alter natural Bengali spelling variations across patient notes such as '□□□□□□□□□□' vs. '□□□□□□□□□' or '□□□□□□' vs. '□□□□□□' and tagged English medical abbreviations embedded in Bengali sentences such as 'MRI', 'ECG', 'CT Scan', 'FNAC', and 'Blood Transfusion' based on their clinical meaning. This diversity makes our corpus much more realistic for training a generalizable model.
- **Handling Category Skew in Round 2 Modeling:** Our EDA clearly shows that patient-facing entities- SYMPTOM and BODY_SITE are much more frequent than clinical prescription entities- MEDICATION, DOSAGE and TEST_RESULT. Knowing this ahead of Round 2 allows Team Pragyan to apply class-weighted loss functions or targeted few-shot prompting with BanglaBERT / IndicBERT so our model achieves high F1 scores on both common and rare medical categories.

Final Submission Verification — Team Pragyan

- Annotated Corpus: Pragyan_Round1_MedBhasha_Annotated.csv (762 notes, 9,172 tokens, BIO format)
- Validation Output: validation_report.txt (Result: PASS | Errors: 0 | Warnings: 0)
- Source Declaration: Included with complete breakdown of open-access & synthetic sources
- EDA Summary: Complete analysis of note lengths, entity frequencies, and co-occurrence patterns